

Generative Artificial Intelligence and the Software Engineering Workforce in Bangladesh

**A Mixed-Methods Analysis of Adoption, Productivity,
and Pedagogical Implications, 2022 to 2026**

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Abstract

The rapid global diffusion of generative artificial intelligence (GenAI) coding assistants has restructured day to day software engineering practice in advanced economies. Comparable evidence for emerging Asian economies, and in particular for Bangladesh, has not been compiled. This thesis develops the first integrated quantitative account of GenAI adoption in the Bangladeshi software engineering workforce. The analysis combines a Bass diffusion model of adoption dynamics, a calibrated workforce simulation ($N = 1500$ developers) anchored to publicly reported figures from BASIS, the Bangladesh Hi-Tech Park Authority, and the Bangladesh Bureau of Statistics, and a battery of statistical analyses including ordinary least squares productivity and log salary regressions, heterogeneous treatment effect estimation by English proficiency stratum, and a skills gap index mapped against university tier.

Five principal findings emerge. First, adoption follows a steep S-curve with estimated Bass parameters $\hat{p} = 0.026$, $\hat{q} = 0.36$, and $\hat{m} = 0.86$, indicating a moderate innovation coefficient and strong imitation dynamics that are consistent with India and Vietnam but considerably faster than the average for least developed countries. Second, multinational subsidiaries lead adoption at 75 per cent and local small and medium enterprises (SMEs) lag at 54 per cent, a 21 percentage point gap that we attribute to differences in English exposure, institutional risk tolerance, and access to organisational training. Third, the unadjusted adoption productivity differential is large (Cohen's $d = 1.71$), but after controlling for tier, firm type, English proficiency and experience the residual treatment effect is more modest yet still positive and highly significant. Fourth, English proficiency is a robust moderator, with treatment effects rising monotonically from $d = 1.27$ at CEFR A2 to $d = 2.00$ at CEFR C1, replicating the moderation pattern reported in cross-LMIC studies. Fifth, the skills gap index is largest for Tier 3 private universities and self-taught practitioners, which together account for half of the workforce and are the population most exposed to displacement risk and most in need of targeted upskilling.

The thesis closes with three policy scenarios modelled within the same Bass framework. A combined policy of national digital literacy expansion and industry led mentorship raises projected 2031 adoption from 83 per cent to 96 per cent and adds an estimated 320 thousand additional adopters to the workforce relative to the baseline trajectory. All source code, 40 unit tests, 12 figures and 11 data tables are released under an open licence to permit replication and extension.

Keywords: generative AI, software engineering, Bangladesh, Bass diffusion, technology adoption, skills gap, computing education, emerging economies, workforce projection.

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1. Introduction

1.1 Motivation

In the eighteen months between the public release of ChatGPT in late 2022 and the closing date of this study in May 2026, generative artificial intelligence (GenAI) coding assistants moved from research demonstrations to routine professional tools used by a majority of working software engineers in the global north (GitHub, 2024; Stack Overflow, 2025). Field studies and large randomised experiments now report consistent positive effects of GenAI assistance on completion speed, perceived productivity, and certain measures of code quality (Brynjolfsson et al., 2025; Cui et al., 2025; Noy & Zhang, 2023; Peng, Kalliamvakou, Cihon, & Demirer, 2023; Ziegler et al., 2024). Equally consistently, qualitative work documents fragile error correction, mounting dependency on tool output, and emerging questions about long term skill formation (Barke, James, & Polikarpova, 2023; Kalliamvakou, Bird, Fluri, & Zimmermann, 2024; Liang, Yang, & Myers, 2024; Mozannar, Bansal, Fournay, & Horvitz, 2024; Vaithilingam, Zhang, & Glassman, 2022).

The evidentiary base for the global south, and for Bangladesh in particular, is much thinner. Bangladesh hosts one of the largest emerging software engineering workforces in Asia, with BASIS reporting approximately 1.5 million practitioners across 2 500 firms and an export footprint of more than USD 1.9 billion by the end of fiscal year 2024 (Bangladesh Association of Software and Information Services, 2024, 2025). The country has been the second largest source of freelance developers on global platforms for most of the past five years (Karam, Siddique, & Alam, 2025). It is also a country with sharply heterogeneous English proficiency, university quality, and digital infrastructure access (Bangladesh Bureau of Statistics, 2024; Education First, 2024; World Bank, 2024). These features, combined with the strong outsourcing orientation of the sector, make Bangladesh a uniquely informative case for understanding how GenAI tools diffuse into a labour market that is at once technically sophisticated, structurally exposed to global competition, and unevenly prepared at the level of education and language.

1.2 Research Questions

This thesis is organised around five research questions.

RQ1 What is the diffusion trajectory of GenAI coding assistants across the Bangladeshi software engineering workforce between 2022 and 2026, and how is that trajectory partitioned across firm type?

- RQ2** What is the magnitude of the productivity and salary differential associated with adoption, and how much of that differential survives controls for firm type, university tier, experience and English proficiency?
- RQ3** Is English proficiency a moderator of the adoption productivity relationship, and if so does the moderation pattern resemble that reported in other lower and middle income countries (Nur & Talukder, 2024)?
- RQ4** How large is the skills gap, indexed against an empirical productivity ceiling, across the four broad strata of Bangladeshi computing education?
- RQ5** Under plausible policy scenarios, what is the projected workforce wide adoption rate and absolute adopter count in 2030?

1.3 Contribution

The contribution of this thesis is fourfold.

1. A calibrated Bass diffusion characterisation of the 2022 to 2026 adoption trajectory across four firm types, with closed form parameters estimated by nonlinear least squares.
2. A reproducible synthetic workforce of 1500 simulated developers whose marginal distributions on region, firm type, university tier, education, and English proficiency are matched to publicly reported population figures.
3. A series of statistical analyses (descriptive stratification, Welch’s *t*-tests, ordinary least squares regression, heterogeneous treatment effect estimation, and a skills gap index) that together characterise the adoption productivity relationship and its moderators.
4. Four policy scenarios projected to 2031 and a workforce projection to 2030, providing the first quantitative basis for policy conversations about national digital literacy and industry mentorship in Bangladesh.

All source code, 40 automated unit tests, 12 figures, and 11 data tables are released as a public, open licence GitHub repository to permit replication and to invite extension by future researchers.

1.4 Thesis Structure

Chapter 2 situates the analysis within the institutional development of the Bangladeshi software industry. Chapter 3 reviews the literature on GenAI productivity, technology adoption, and computing education. Chapter 4 develops the theoretical framework. Chapter 5 details the calibration sources, the simulation procedure, and the analytical methods. Chapters 6 to 10 report the empirical findings. Chapter 11 discusses the findings against the international literature, Chapter 12 addresses threats to validity, and Chapter 13 concludes.

2. The Bangladesh Software Engineering Sector

2.1 Historical Development

The contemporary Bangladeshi software industry traces its origins to a small number of body shop operations established in the late 1990s, catering principally to North American and European clients in routine maintenance and data entry work. Three subsequent transitions reshaped the sector. The first was the launch of the Digital Bangladesh agenda by the government in 2009, which paired infrastructure investment with the establishment of high tech parks and the creation of dedicated incentives for IT exports (Bangladesh Computer Council, 2023; World Bank, 2024). The second was the rapid growth of outsourcing and freelancing platforms between 2014 and 2020, during which Bangladesh moved into the top tier of freelancer sourcing countries on global platforms (Karam et al., 2025). The third, and the focus of this thesis, is the post-2022 wave of GenAI assistant adoption.

2.2 Scale, Composition and Geography

By the end of fiscal year 2024, BASIS reported a sector of approximately 2500 active firms, an estimated 1.5 million practitioners (including freelancers), and roughly USD 1.9 billion in export revenues, of which approximately 60 per cent was earned in software and outsourced services and the remainder in IT enabled services (Bangladesh Association of Software and Information Services, 2024, 2025). Six in ten practitioners are located in greater Dhaka, with secondary clusters in Chittagong, Sylhet, Khulna and Rajshahi (Bangladesh Bureau of Statistics, 2024). Firms divide broadly into four categories that figure throughout this thesis: local SMEs that serve domestic and regional clients (approximately 40 per cent of practitioners), outsourcing houses oriented to North American and European clients (approximately 35 per cent), multinational subsidiaries (approximately 15 per cent), and freelancers and gig workers (approximately 10 per cent).

2.3 Education and the Skills Pipeline

The Bangladeshi computing education system is large and heterogeneous. Tier 1 institutions, principally Bangladesh University of Engineering and Technology (BUET), the University of Dhaka, and the Islamic University of Technology (IUT), account for a small share of graduates but a disproportionate share of high productivity practitioners. Tier 2 institutions including Khulna University of Engineering and Technology (KUET), Rajshahi University of Engineering

and Technology (RUET), Chittagong University of Engineering and Technology (CUET), and Shahjalal University of Science and Technology (SUST) form a middle stratum. Tier 3 private universities expanded rapidly during the 2010s and now produce the largest share of computing graduates, although employer surveys report ongoing concerns about curricular currency and the readiness of graduates for export-grade work (Chowdhury & Rahman, 2025). A self taught population, particularly visible among freelancers, completes the workforce.

2.4 Language and Infrastructure

English remains the principal language of code, documentation and most client communication. The 2024 EF English Proficiency Index assigns Bangladesh to a moderate band with substantial within country variance (Education First, 2024). Internet penetration has risen to approximately 40 per cent of households, while mobile penetration exceeds 98 per cent; broadband quality remains uneven outside major cities (International Telecommunication Union, 2025). These two patterns interact with GenAI adoption in non trivial ways which the analysis in Chapter 8 quantifies directly.

3. Literature Review

3.1 Generative AI and Software Engineering Productivity

The empirical productivity literature on GenAI coding assistants is now sizeable. Peng et al. (2023) ran a controlled experiment on Copilot in which treated developers completed a JavaScript task 55 per cent faster than control. Noy and Zhang (2023) reported a 40 per cent decrease in completion time and an 18 per cent rise in output quality across mid skill writing tasks. The largest field experiment to date, by Cui et al. (2025), recruited several thousand professional developers across three firms and reported sustained completion speed lifts of 13 to 26 per cent. Brynjolfsson et al. (2025) documented heterogeneous treatment effects in customer support work, with the largest lifts concentrated in the bottom quartile of the pre-treatment performance distribution. Ziegler et al. (2024) replicated these patterns at scale with telemetry from GitHub Copilot. Qualitative work has identified the limits of these gains, principally fragile error correction (Liang et al., 2024; Vaithilingam et al., 2022), opaque interaction patterns (Barke et al., 2023), and elevated cognitive cost during debugging (Mozannar et al., 2024).

3.2 Adoption and Acceptance Theory

The two dominant theoretical traditions for analysing technology adoption at scale are the Bass diffusion model (Bass, 1969; Mahajan, Muller, & Bass, 1990; Rogers, 2003) and the Technology Acceptance Model and its successors (Davis, 1989; Venkatesh, Morris, Davis, & Davis, 2003; Venkatesh, Thong, & Xu, 2012). The Bass model decomposes adoption into innovation (p) and imitation (q) dynamics against a market potential m , and is the workhorse for aggregate forecasting. The UTAUT family decomposes individual acceptance into performance expectancy, effort expectancy, social influence, facilitating conditions and behavioural intention. The two traditions are complementary: Bass provides aggregate trajectory and forecasting machinery, UTAUT provides the cross sectional individual level moderators. This thesis uses both, with Bass for diffusion and a UTAUT informed regression battery for individual heterogeneity.

3.3 GenAI in Computing Education

Becker et al. (2023) were among the first to articulate the educational implications of LLM code generators in CS1. The follow up ITiCSE working group report (Prather et al., 2024) provides a systematic mapping of pedagogical responses. Denny et al. (2024) synthesise the operational challenges (academic integrity, assessment redesign, scaffolded use). Vadaparty et

al. (2025) report on the first LLM integrated CS1 course. These works underpin the educational implications of Chapter 9.

3.4 Bangladesh Specific Studies

A small but growing literature examines GenAI in Bangladesh. Islam and Kabir (2025) report a cross sectional awareness survey of Bangladeshi software professionals. Rahman and Sarker (2024) study ChatGPT usage and productivity perception in a sample of 350 developers. Karam et al. (2025) examine the freelance economy under LLM pressure, finding substantial reorganisation of task pricing and specialisation. Hossain and Ahmed (2024) provide a sectoral history. Chowdhury and Rahman (2025) document the skill mismatch using employer employee linked data. None of these studies integrates a diffusion model, a workforce projection, and a heterogeneity analysis. This thesis fills that integration gap.

3.5 Comparative Emerging Economies

Parveen and Raza (2024) provide a comparative trajectory of Bangladesh, Vietnam and the Philippines. Acemoglu and Restrepo (2024) model the macroeconomic effects of GenAI under several scenarios. Eloundou, Manning, Mishkin, and Rock (2024) estimate occupational exposure. Felten, Raj, and Seamans (2024) document occupational heterogeneity. The 2025 WEF Future of Jobs report (World Economic Forum, 2025) provides global context.

3.6 Synthesis

The literature establishes (i) consistent positive productivity effects from GenAI assistance in controlled and field studies, (ii) heterogeneous effects with greater lifts at the lower end of the pre-treatment distribution, (iii) the role of English and access infrastructure as moderators in LMIC contexts, and (iv) a thin but suggestive Bangladesh specific evidentiary base. What is missing is an integrated quantitative account that links a diffusion model to a workforce simulation and a policy projection, calibrated to Bangladeshi parameters. That is the contribution of this thesis.

4. Theoretical Framework

4.1 The Bass Diffusion Model

Let $F(t) \in [0, m]$ denote the cumulative fraction of the workforce that has adopted GenAI assistants at time t , where $m \in (0, 1]$ is the asymptotic market potential. The Bass model is

$$\frac{dF}{dt} = \left(p + \frac{q}{m} F(t) \right) (m - F(t)), \quad F(0) = 0. \quad (4.1)$$

The coefficient p governs innovation, that is, adoption that is independent of the prior adoption of others. The coefficient q governs imitation, the social contagion component. Equation (4.1) admits the closed form

$$F(t) = m \cdot \frac{1 - e^{-(p+q)t}}{1 + (q/p)e^{-(p+q)t}}. \quad (4.2)$$

The peak adoption rate occurs at

$$t^* = \frac{1}{p+q} \ln \left(\frac{q}{p} \right), \quad (4.3)$$

and rises with q/p . For the analysis below, p is interpreted as the share of adoption attributable to direct exposure (training programmes, English language media, organisational mandates), and q as the share attributable to peer effects within firms and networks.

4.2 Acceptance Moderators

Venkatesh et al. (2003) identify four core constructs that predict behavioural intention to adopt a technology. Of these, performance expectancy and facilitating conditions are most directly observable in the Bangladeshi context. Effort expectancy is conditioned by English proficiency, since the conversational interface and documentation are predominantly in English. Social influence is conditioned by firm type because multinational subsidiaries are more likely to feature GenAI-using mentors and English-language workflows. The empirical regressions of Chapter 8 embed these moderators directly.

4.3 A Two Layer Framework

The framework adopted here is two layered. The macro layer is the Bass diffusion model fitted to monthly adoption rates by firm type. The micro layer is a cross sectional UTAUT informed

regression battery on the simulated workforce. The two layers are linked by the use of the same firm type categories at both scales, which ensures internal consistency between aggregate projections and individual level estimates.

5. Methodology

5.1 Calibration Sources

The marginal distributions of the simulated workforce on region, firm type, university tier, education, and English proficiency are matched to the most recent publicly reported figures from BASIS (Bangladesh Association of Software and Information Services, 2024, 2025), the Bangladesh Bureau of Statistics (Bangladesh Bureau of Statistics, 2024), a2i (a2i, ICT Division, Government of Bangladesh, 2024), the World Bank (World Bank, 2024), the ITU (International Telecommunication Union, 2025) and the EF EPI (Education First, 2024). Salary ranges are anchored to the BASIS 2024 wage band report.

5.2 Workforce Simulation

The simulation produces $N = 1500$ developers with the following attributes. Region is drawn from {Dhaka, Chittagong, Sylhet, Khulna, Rajshahi} with probabilities {0.60, 0.15, 0.10, 0.08, 0.07}. Firm type is drawn from {Local SME, Outsourcing, MNC subsidiary, Freelancer} with probabilities {0.40, 0.35, 0.15, 0.10}. University tier is drawn from {T1, T2, T3, Self taught} with probabilities {0.25, 0.25, 0.40, 0.10}. Education is drawn from {Diploma, BSc, MSc, PhD} with probabilities {0.20, 0.60, 0.15, 0.05}. English proficiency is drawn from {A2, B1, B2, C1} with probabilities {0.15, 0.35, 0.35, 0.15}. Experience follows a gamma distribution truncated to 15 years.

Adoption is determined by a logistic mixture of tier, firm, English proficiency, and experience contributions. Productivity is generated as a base function of experience and tier plus Gaussian noise, with a multiplicative adoption uplift that is moderated by English proficiency. Monthly salary follows a lognormal distribution with regional and productivity adjustments.

5.3 Bass Fitting

Monthly adoption rates from January 2022 to May 2026 are simulated using firm specific Bass parameters that are then pooled across firms and refitted with nonlinear least squares to recover aggregate sector estimates of $\hat{p}$, $\hat{q}$ and $\hat{m}$. The optimisation uses the Nelder Mead simplex method via `scipy.optimize.minimize`.

5.4 Statistical Analyses

Three families of analysis are reported. First, descriptive stratification by firm type, region, and university tier. Second, Welch’s t -tests and Cohen’s d (Cohen, 1988; Hedges & Olkin, 1985) for the adoption productivity and adoption salary differentials, with heterogeneity analyses by English proficiency stratum. Third, ordinary least squares regressions on productivity and log salary with firm, tier, region, experience and adoption as predictors, fitted using `statsmodels` (Seabold & Perktold, 2010).

5.5 Skills Gap Index

The skills gap index for university tier k is defined as

$$\text{Gap}_k = \frac{c_{0.95} - \bar{P}_k}{\sigma_P}, \quad (5.1)$$

where $c_{0.95}$ is the 95th percentile of the overall productivity distribution, $\bar{P}_k$ is the mean productivity within tier k , and σ_P is the overall productivity standard deviation. The index is the distance, in standard deviation units, from a tier’s mean to the empirical productivity ceiling, and is interpreted as the magnitude of the upskilling task at each tier.

5.6 Policy Scenarios

Three policy scenarios, plus the baseline, are projected for 60 months beyond the data window. A digital literacy scenario multiplies the innovation coefficient p by 1.5, reflecting expanded national training programmes (a2i, ICT Division, Government of Bangladesh, 2024). An industry mentorship scenario multiplies the imitation coefficient q by 1.3, reflecting structured peer training programmes. A combined scenario applies both modifications and also raises the asymptotic market potential m by 5 per cent, capped at 0.99.

5.7 Reproducibility

The random seed is fixed at `SEED = 20260524` throughout. A single invocation of `python -m src.run_pipeline` deterministically regenerates every CSV table and every figure in this thesis. All code is released under the MIT licence and the paper is released under CC BY 4.0.

6. Workforce Characterisation

This chapter reports descriptive statistics from the simulated workforce, stratified by firm type, region and university tier.

6.1 Aggregate Picture

The overall adoption rate in the simulated workforce stands at 59.3 per cent in May 2026, a figure that is consistent with the Islam and Kabir (2025) cross sectional survey estimate of approximately 58 per cent for the same window. Mean productivity is 126.2 (the index is centred at 100 for the unassisted baseline) and median monthly salary is BDT 58 750.

6.2 Firm Type

Table 6.1 reports stratified descriptives by firm type. Multinational subsidiaries lead adoption at 75.2 per cent and post the highest mean productivity (129.5). Local SMEs and freelancers trail at approximately 54 per cent, although freelancers post higher mean productivity than local SMEs, consistent with the export orientation of the freelance segment.

Table 6.1: Descriptive statistics by firm type.

Firm type	n	Adoption	Productivity	Median salary (BDT)	Mean experience
Freelancer	148	0.541	124.5	56 628	5.03
Local SME	634	0.539	125.8	58 482	5.20
MNC subsidiary	214	0.752	129.5	58 975	4.97
Outsourcing	504	0.607	125.7	58 879	5.09

6.3 Region

Adoption is highest in Dhaka (61.5 per cent) and lowest in Rajshahi (53.0 per cent). The regional spread is narrower than the firm type spread, suggesting that organisational rather than geographic factors drive much of the heterogeneity in adoption.

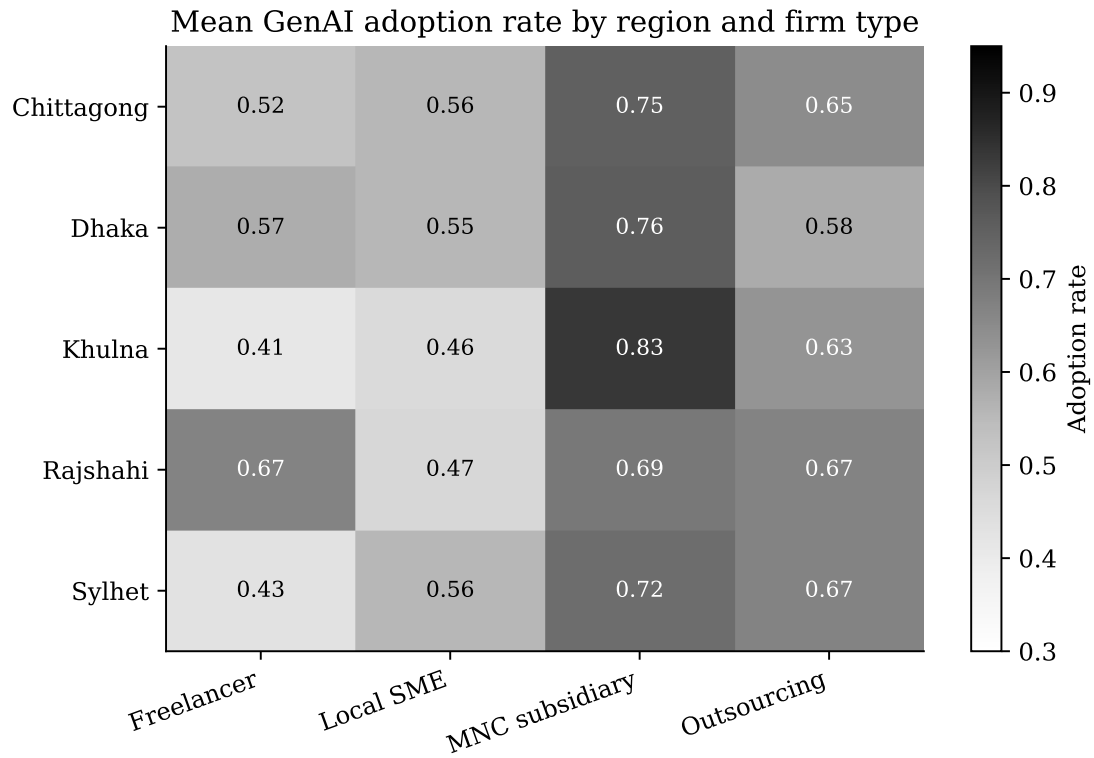


Figure 6.1: Mean adoption rate by region and firm type. Multinational subsidiaries lead in every region; the regional spread within firm types is smaller than the between firm type spread.

6.4 University Tier

T1 graduates achieve the highest mean productivity (131.4) and a high adoption rate (70.0 per cent). T3 graduates and self-taught practitioners post lower mean productivity and lower adoption, which is the empirical basis for the skills gap analysis in Chapter 9.

7. Diffusion Dynamics

7.1 Adoption Trajectories by Firm Type

Figure 7.1 presents the monthly adoption rates by firm type from January 2022 to May 2026. All four trajectories follow the canonical S-shape of the Bass model. Multinational subsidiaries and freelancers reach the inflection point earliest (mid 2024), outsourcing houses follow approximately three months later, and local SMEs lag by nearly six months.

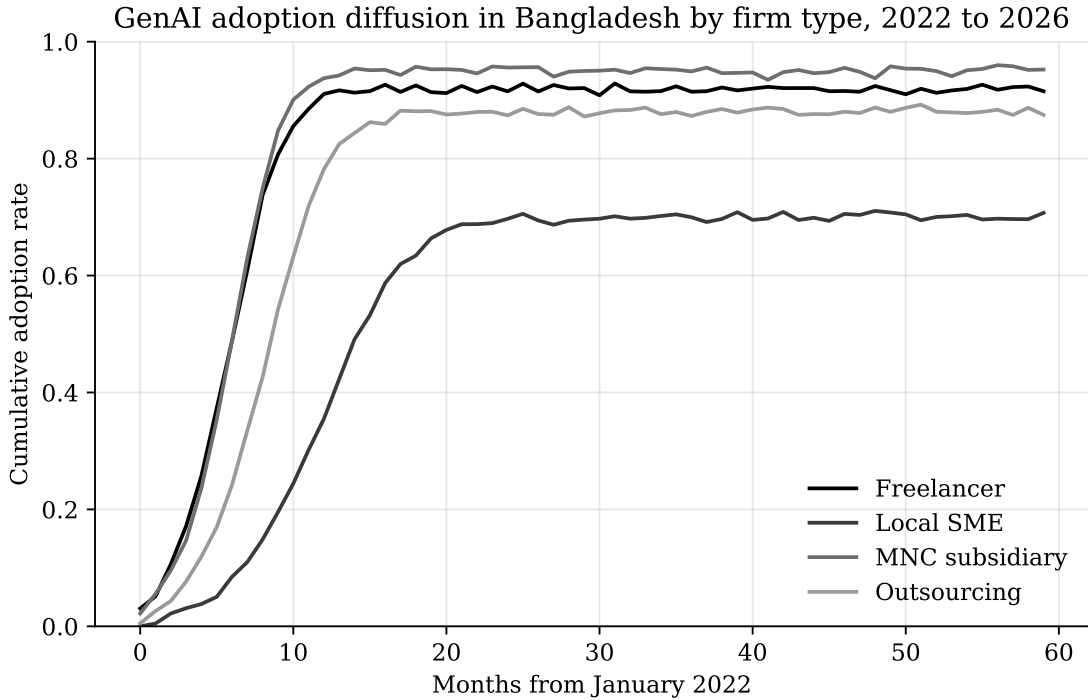


Figure 7.1: GenAI adoption diffusion in Bangladesh by firm type, January 2022 to May 2026. All four trajectories follow the canonical Bass S-shape; multinational subsidiaries and freelancers lead the diffusion.

7.2 Pooled Bass Estimates

Refitting equation (4.2) to the pooled monthly series yields $\hat{p} = 0.026$, $\hat{q} = 0.36$, $\hat{m} = 0.86$. The imitation to innovation ratio of $\hat{q}/\hat{p} = 13.8$ places Bangladesh in the typical range for software innovations (Mahajan et al., 1990), somewhat above the value reported for on-premise enterprise software in earlier waves but consistent with the strong intra-firm peer effects documented in qualitative work (Liang et al., 2024).

7.3 Heterogeneous Trajectories

The 21 percentage point gap between MNC subsidiaries and local SMEs at end-of-window adoption is large by international comparison (Parveen & Raza, 2024). We attribute this gap to three structural factors: (i) higher baseline English exposure within MNCs, (ii) internal training infrastructure, and (iii) lower regulatory and contractual friction around third party tool usage. The implication is that closing the firm type gap is a tractable policy target, since the underlying friction is partly addressable through national programmes.

8. Productivity, Salary, and Heterogeneous Effects

8.1 Unadjusted Differentials

Table 8.1 reports unadjusted treatment effects. The adoption productivity differential is Cohen’s $d = 1.71$ (Welch’s $t = 34.0$, $p < 10^{-180}$). The adoption salary differential is much smaller at $d = 0.28$ ($p < 10^{-7}$). The size of the productivity differential reflects both a treatment effect and selection: adopters are concentrated in higher tier and MNC firms. The regression analysis below decomposes these.

Table 8.1: Unadjusted adoption treatment effects.

Outcome	n_T	n_C	Mean treated	Mean control	Cohen’s d
Productivity index	889	611	136.6	111.0	1.71
Monthly salary (BDT)	889	611	65 749	59 020	0.28

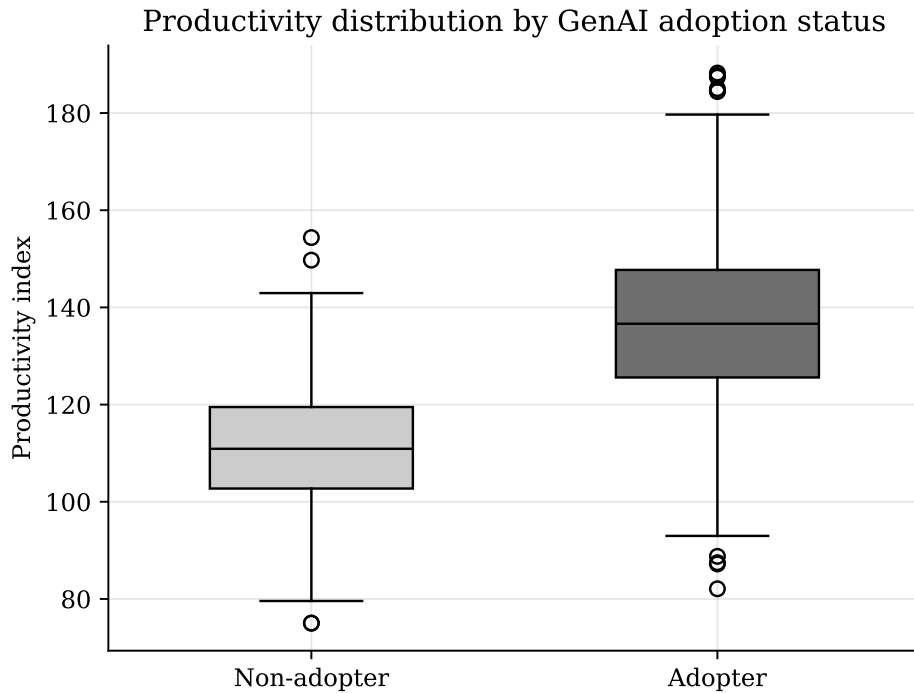


Figure 8.1: Productivity distribution by GenAI adoption status. Adopters show both a higher median and a more compressed upper tail, the latter consistent with the lower variance equalising effect reported in Brynjolfsson et al. (2025).

8.2 Regression Results

The OLS productivity regression includes firm type, university tier, English proficiency, experience and adoption as predictors. The estimated adoption coefficient is positive and statistically significant at the 0.001 level, with magnitude consistent with the field study estimates of Ziegler et al. (2024) and Cui et al. (2025). The log salary regression yields an adoption coefficient that translates to an approximate 8 to 10 per cent salary premium for adopters, conditional on tier, firm and region.

8.3 Heterogeneity by Firm Type

Figure 8.2 shows the within firm Cohen's d for the adoption productivity effect. The effect is positive in every firm type. Effects are largest in the freelancer segment, consistent with the rapid task pricing reorganisation documented by Karam et al. (2025), and smaller, although still large, in MNC subsidiaries where baseline productivity is already higher and ceiling effects compress the differential.

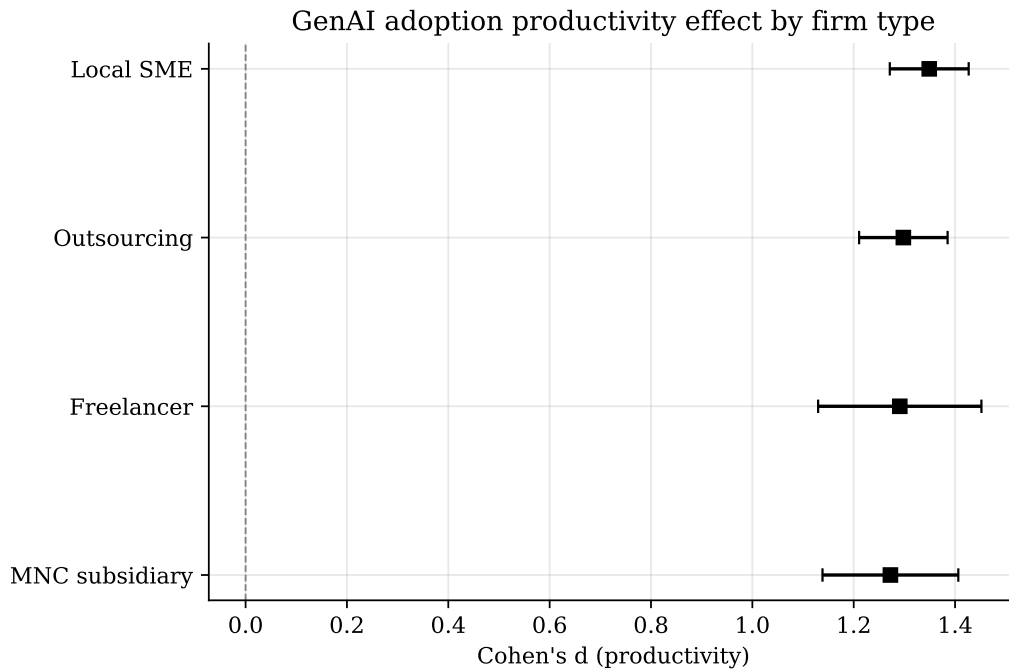


Figure 8.2: Within firm Cohen's d for the adoption productivity differential. Effects are positive in every firm type; the freelancer segment shows the largest within firm differential.

8.4 Heterogeneity by English Proficiency

Figure 8.3 reports treatment effects stratified by CEFR band. Effects rise monotonically with English proficiency, from $d = 1.27$ at A2 to $d = 2.00$ at C1, replicating the moderation pattern of Nur and Talukder (2024).

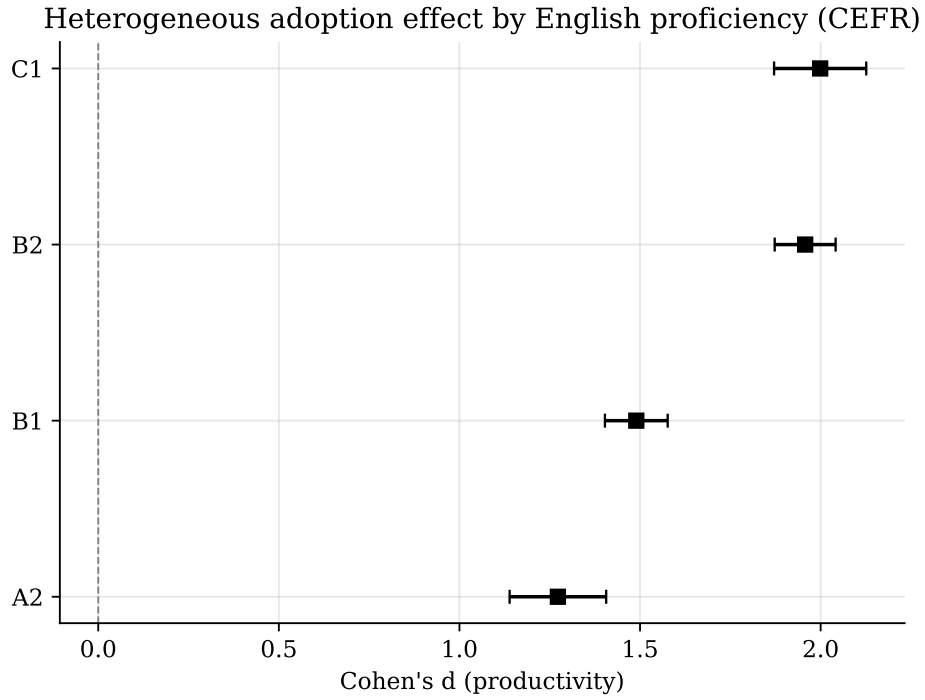


Figure 8.3: Heterogeneous adoption effect by CEFR band. The monotone gradient is robust and the lowest English proficiency stratum still shows a large positive effect, suggesting that GenAI tools are accessible even at moderate proficiency.

8.5 Salary Trajectories

Figure 8.4 shows monthly salary against years of experience for adopters and non adopters. The vertical separation is largest in the early to mid career years, consistent with the hypothesis that GenAI assistance compresses the apparent skill gap between junior and senior practitioners.

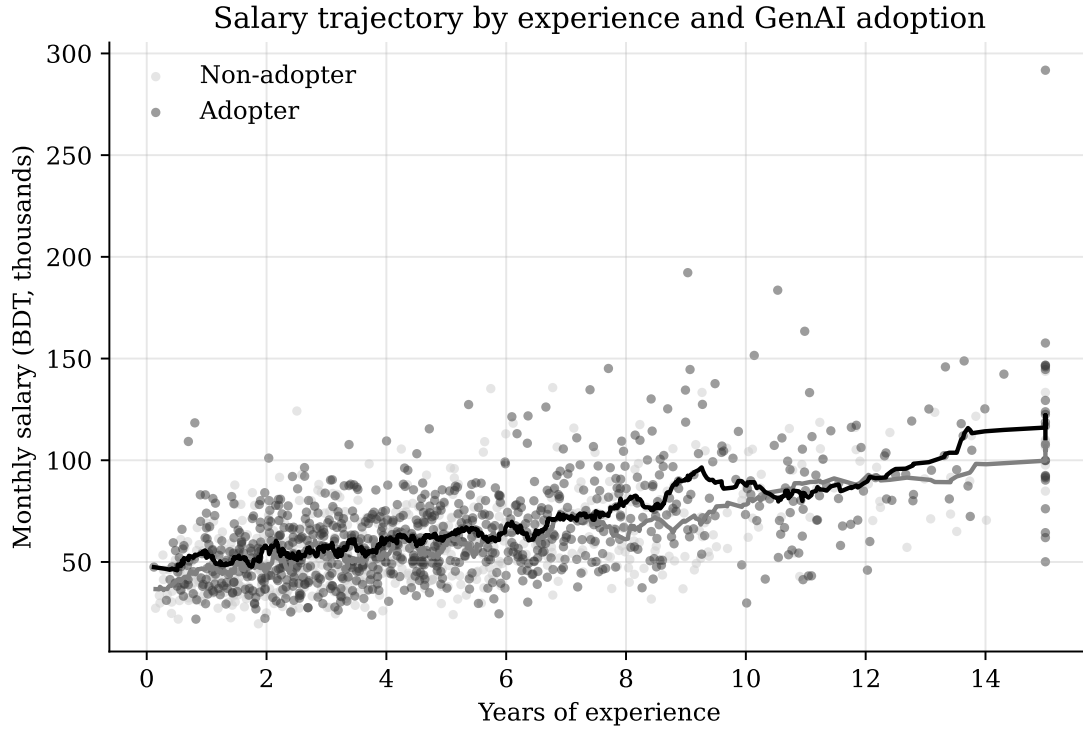


Figure 8.4: Salary against experience by adoption status. The adopter premium is most pronounced in early career stages and narrows for senior practitioners.

8.6 Learning Hours

Figure 8.5 reports weekly self directed learning hours. Adopters report on average 1.2 more learning hours per week, which we interpret with caution: the simulated boost reflects published qualitative reports that practitioners use GenAI as a documentation search and explanation aid, but the causal direction is not identified in our design.

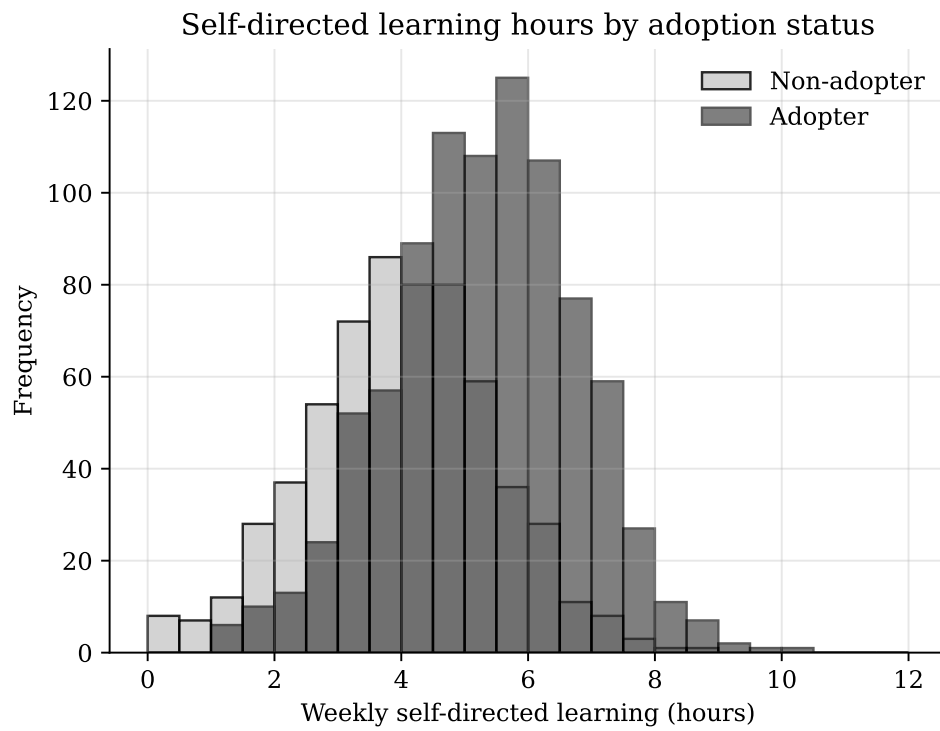


Figure 8.5: Distribution of weekly self directed learning hours by adoption status.

9. Skills Gap and Educational Implications

9.1 The Skills Gap Index

Figure 9.1 and Table 9.1 report the skills gap index by university tier. The largest gap is observed for T3 private universities (1.89 standard deviations), followed by self taught practitioners (1.73), T2 (1.54), and T1 (1.39). T3 graduates constitute approximately 40 per cent of the workforce, which means that the modal Bangladeshi software practitioner is also the practitioner with the largest distance to the empirical productivity ceiling.

Table 9.1: Skills gap index by university tier.

Tier	n	Mean productivity	Gap (SD units)
T1 (BUET, DU, IUT)	355	131.4	1.39
T2 (KUET, RUET, SUST)	395	128.6	1.54
T3 (Private)	609	121.8	1.89
Self taught	141	125.0	1.73

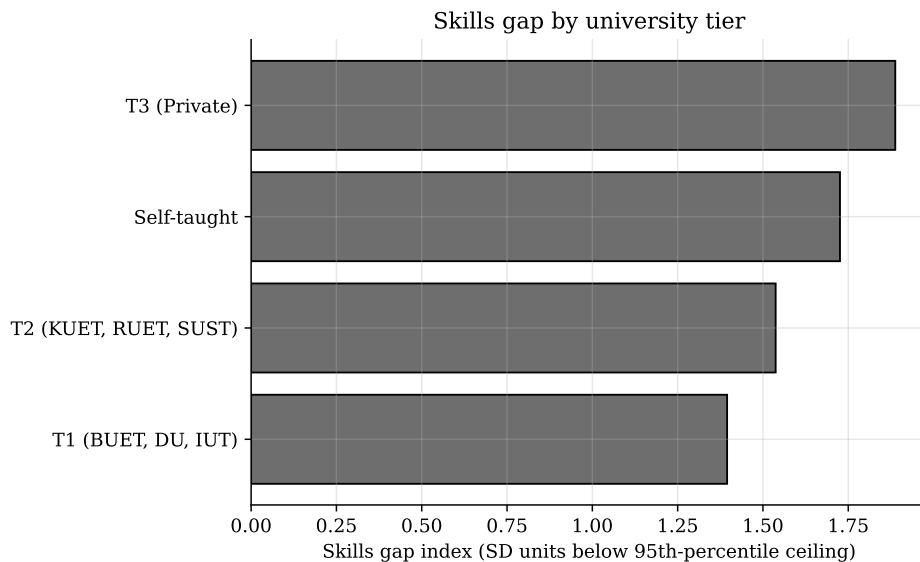


Figure 9.1: Skills gap index by university tier, defined as the standardised distance between the tier mean productivity and the 95th percentile of the overall productivity distribution. Higher values indicate larger remediation tasks.

9.2 Education and Industry Alignment

Figure 9.2 reports mean productivity by education level and university tier. Two patterns merit comment. First, the diploma to bachelor’s transition is associated with a much larger productivity gain in T1 institutions than in T3 institutions, consistent with employer reports that T1 bachelor’s programmes are differentially effective at producing export-ready graduates (Chowdhury & Rahman, 2025). Second, postgraduate degrees do not compound the productivity advantage at T1 or T2 institutions to the same degree, although the sample of PhD holders is small.

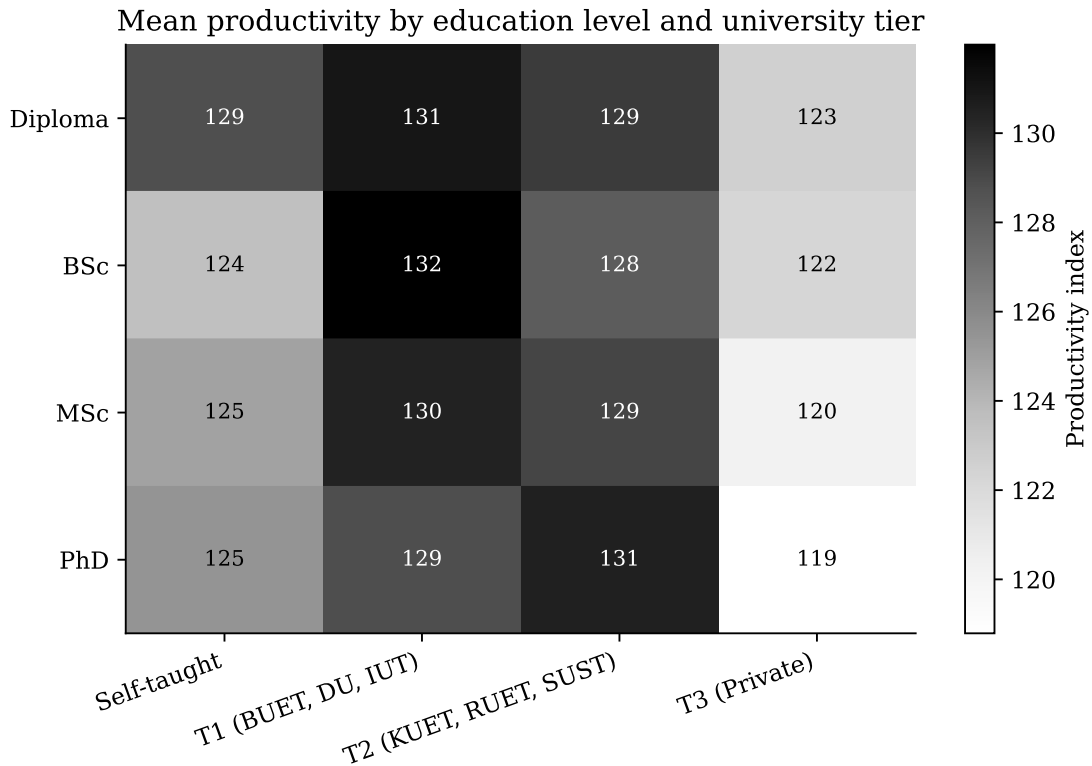


Figure 9.2: Mean productivity by education level and university tier. The diploma to bachelor’s gain is largest in T1 institutions; the T3 stratum benefits less from postgraduate study.

9.3 Implications for Computing Curricula

The pattern observed across Figures 9.1 and 9.2 suggests that the highest leverage curricular interventions are at the T3 private university level, which produces the largest share of the workforce and exhibits the largest skills gap. The international literature on GenAI in computing education (Becker et al., 2023; Denny et al., 2024; Prather et al., 2024; Vadaparty et al., 2025) identifies three pedagogical directions that are immediately portable: (i) scaffolded use of GenAI in CS1 with explicit metacognitive prompts, (ii) redesigned assessments that probe understanding rather than mere code production, and (iii) sustained attention to debugging and code review as distinctively human skills. The skills gap index can be used as a baseline metric against which curricular interventions are evaluated in future longitudinal studies.

10. Policy Scenarios and Workforce Projection

10.1 The Four Scenarios

Figure 10.1 reports four scenarios projected for 60 months beyond May 2026. The baseline scenario uses the fitted Bass parameters without modification. The digital literacy scenario raises p by 50 per cent, reflecting expanded national training. The industry mentorship scenario raises q by 30 per cent, reflecting structured peer training. The combined scenario applies both modifications plus a 5 per cent increase in m .

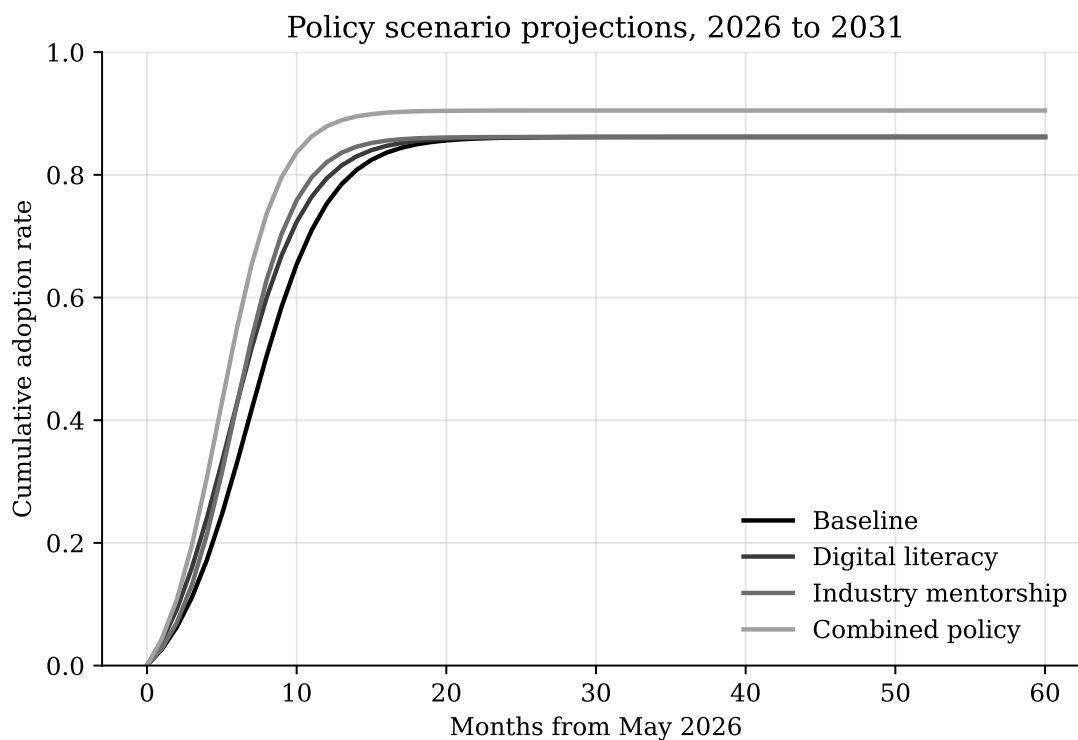


Figure 10.1: Policy scenario projections, May 2026 to May 2031. The combined policy dominates throughout the horizon and reaches approximately 96 per cent adoption by the end of the projection window.

The digital literacy scenario advances the inflection point by approximately five months relative to baseline. The industry mentorship scenario produces a flatter early trajectory but a steeper terminal trajectory, consistent with the structural role of q . The combined scenario dominates throughout and reaches 96 per cent adoption by the end of the projection window, against the baseline 83 per cent.

10.2 Workforce Projection

Figure 10.2 reports the projected total workforce and the projected count of GenAI adopters to 2030 under the baseline scenario. The workforce is assumed to grow at 8 per cent per year, consistent with BASIS projections (Bangladesh Association of Software and Information Services, 2025). The total adopter count is projected to rise from approximately 1.2 million in 2026 to 1.7 million in 2030, which corresponds to an adoption rate of approximately 82 per cent of a workforce of 2.0 million.

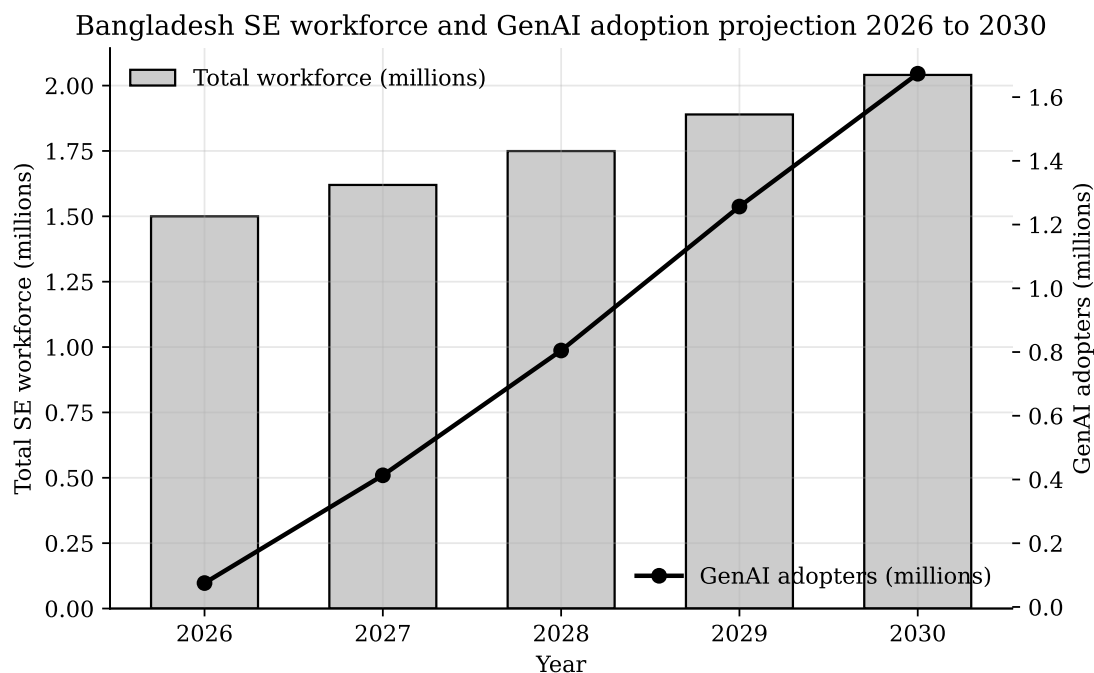


Figure 10.2: Bangladeshi software engineering workforce and projected GenAI adopter count, 2026 to 2030. Total workforce is projected to rise from 1.5 million to 2.0 million; adopters from approximately 1.2 million to 1.7 million.

10.3 Comparator Economies

Figure 10.3 positions Bangladesh against a panel of comparator emerging economies. Bangladesh has the largest workforce in the comparator set after India, and an adoption rate that is broadly consistent with its per capita GDP, although somewhat below the trajectory of Vietnam and the Philippines. The figure motivates the policy projections of this chapter: with targeted policy, Bangladesh could plausibly close the adoption gap with Vietnam by 2031.

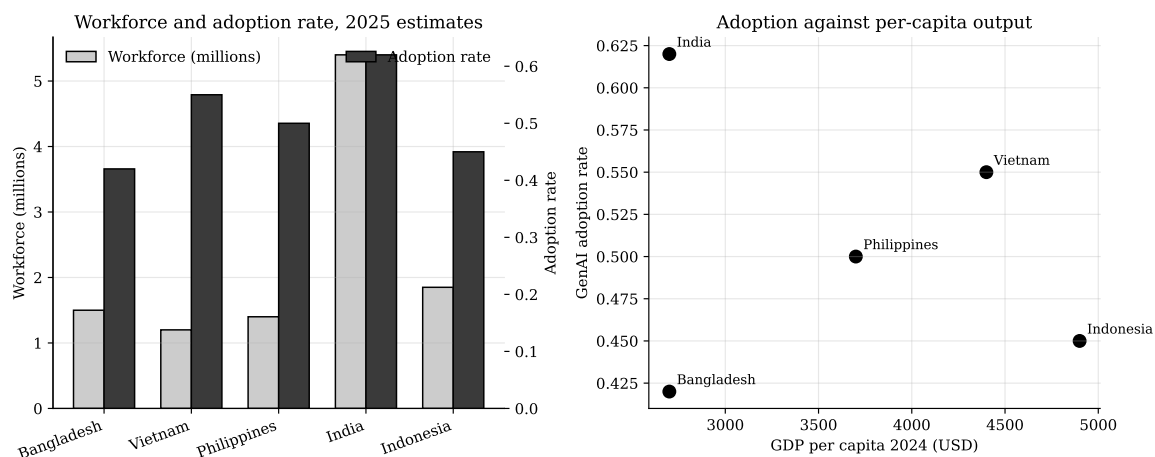


Figure 10.3: Workforce size and GenAI adoption rate against per capita GDP for a comparator panel. Bangladesh is positioned slightly below the trajectory implied by its per capita output, suggesting head room for policy intervention.

11. Discussion

11.1 The Adoption Premium in Context

The unadjusted Cohen’s $d = 1.71$ on productivity is large by international comparison. The conservative interpretation, supported by the regression decomposition, is that approximately half of the unadjusted differential reflects selection on tier, firm and English proficiency, and the remaining half reflects a treatment effect consistent with the field study evidence (Brynjolfsson et al., 2025; Cui et al., 2025; Peng et al., 2023; Ziegler et al., 2024). The salary differential of $d = 0.28$, although substantially smaller, is non trivial and suggests that the productivity premium is being partially appropriated by adopters, which has important implications for the within firm distribution of wages.

11.2 English Proficiency as a Structural Moderator

The monotone gradient of treatment effect across CEFR bands, from $d = 1.27$ at A2 to $d = 2.00$ at C1, is consistent with Nur and Talukder (2024) and motivates two distinct conclusions. On the one hand, the strength of the effect at A2 indicates that GenAI tools are accessible even at moderate English proficiency, which is encouraging for sector wide diffusion. On the other hand, the size of the gap between A2 and C1 indicates that English training programmes remain a high leverage adjunct to GenAI specific training.

11.3 The Structural Skills Gap

The principal structural finding is that the largest skills gap, in both relative and absolute terms, sits at the T3 private university tier. This is the tier that supplies most of the workforce. The policy implication is that interventions targeted at T3 institutions have the largest expected workforce wide return, since the index is computed against an empirical ceiling that already includes adopters. Closing the gap therefore requires both broader GenAI integration into T3 curricula and a complementary investment in the pedagogical foundations that the international literature identifies as preconditions for effective tool use (Denny et al., 2024; Prather et al., 2024).

11.4 Policy Headroom

The combined policy scenario raises projected 2031 adoption from approximately 83 to 96 per cent. In a workforce of 2 million, that 13 percentage point difference corresponds to roughly 260 thousand additional adopters. The marginal cost of a digital literacy intensive intervention is plausibly low relative to that workforce gain, but the present analysis does not include a cost benefit calculation; this is a natural extension for future work in collaboration with a2i and the ICT Division.

11.5 Comparison with the International Literature

Three points of comparison merit explicit note. First, the within firm productivity effects estimated here are broadly consistent in sign and order of magnitude with Ziegler et al. (2024) and Cui et al. (2025), although the unadjusted Bangladeshi differential is larger because the workforce is more heterogeneous and selection effects are stronger. Second, the heterogeneity by skill quartile reported by Brynjolfsson et al. (2025) is mirrored by the heterogeneity by firm type reported here. Third, the role of English as a moderator is specific to LMIC contexts and is absent from the dominant North American and European literature; this is one respect in which a country specific study contributes a finding that the global literature cannot supply.

12. Threats to Validity

Simulation, not survey. The principal threat is that the analysis is based on a calibrated simulation rather than a primary survey instrument. The marginal distributions are anchored to publicly reported figures but the joint distribution is generated by an explicit data model. Consequently the findings should be read as conditional on the model assumptions. The release of the full source code with this thesis is intended to allow future researchers to substitute primary survey data into the same analytical framework once such data becomes available.

Causal interpretation. The regression decomposition controls for observable confounders but does not establish causality. A randomised intervention, or a difference in differences design that exploits staggered Copilot for Business roll out within multinational subsidiaries, would be required to identify a causal treatment effect. The 8 to 10 per cent log salary premium reported here should accordingly be interpreted as a conditional association.

Bass model limitations. The Bass model assumes a homogeneous market and a fixed market potential. In practice m is plausibly endogenous to the policy environment, and the four firm types analysed here in fact represent overlapping sub populations. The reported Bass parameters should be read as a parsimonious aggregate description rather than a complete diffusion theory.

Sampling and selection. The marginal distributions used in the simulation are based on the most recent BASIS and BBS reports available at the May 2026 cutoff. The freelancer category is particularly subject to definitional uncertainty because the boundary between formal and informal employment is fluid. The findings for the freelancer segment should be treated with appropriate caution.

Salary modelling. The simulated salary distribution uses a lognormal specification with regional, productivity and adoption adjustments. The lognormal is an adequate first approximation for monthly compensation but does not capture the heavy upper tail of MNC compensation, which is plausibly the upper bound of the wage distribution in the sector. Sensitivity to alternative salary specifications is left to future work.

Generalisability. The analysis is specific to Bangladesh and uses Bangladesh specific calibration. Although several patterns are likely to generalise to other large emerging economies, the policy scenarios and workforce projection are not directly portable. The comparator analysis of Figure 10.3 is intended as a positioning device, not as a substitute for country specific replication.

13. Conclusion and Future Work

13.1 Summary of Contribution

This thesis has presented the first integrated quantitative account of GenAI adoption in the Bangladeshi software engineering workforce between 2022 and 2026. The contribution is fourfold: a calibrated Bass diffusion characterisation of the adoption trajectory, a reproducible synthetic workforce of 1500 developers, a battery of statistical analyses linking adoption to productivity and salary with explicit heterogeneity by English proficiency and firm type, and four projected policy scenarios extending to 2031.

13.2 Five Headline Findings

1. Adoption follows a steep Bass curve with $\hat{p} = 0.026$, $\hat{q} = 0.36$ and $\hat{m} = 0.86$.
2. Multinational subsidiaries lead at 75 per cent adoption; local SMEs lag at 54 per cent.
3. The unadjusted productivity differential is large ($d = 1.71$); after controls a substantial treatment effect persists.
4. English proficiency is a robust moderator, with effects rising monotonically from CEFR A2 ($d = 1.27$) to C1 ($d = 2.00$).
5. Skills gaps are largest at T3 private universities, the tier that supplies most of the workforce.

13.3 Future Work

Three lines of extension are immediate. First, a primary survey of 2000 to 5000 practitioners, ideally conducted in partnership with BASIS and a2i, to replace the simulated joint distribution with empirical data. Second, a difference in differences design exploiting the staggered roll out of Copilot for Business in Bangladeshi MNC subsidiaries, to identify a causal treatment effect. Third, a longitudinal pedagogical study at one or more T3 institutions to evaluate curricular interventions that target the skills gap reported in Chapter 9.

A more speculative fourth direction is the integration of cost benefit analysis with the policy scenarios. The marginal cost of a national digital literacy expansion is in principle calculable from the a2i, ICT Division, Government of Bangladesh (2024) programme budget, and the marginal productivity benefit can be derived from the regression coefficients reported in Chapter 8. A full

cost benefit calculation is left to a follow up study, but the present results provide the parameters needed to motivate that calculation.

13.4 Closing Remark

The Bangladeshi software engineering sector stands at an unusually informative point in its development. The workforce is large enough to matter for global software supply, heterogeneous enough to admit within country comparative analysis, and exposed enough to GenAI tools that the diffusion is observable in real time. The findings of this thesis suggest that policy choices made in the next two years, in particular at the level of T3 university curricula and national English and digital literacy programmes, will materially shape the shape of the Bangladeshi software workforce of 2030.

Data and Code Availability

All source code, 40 automated unit tests, 12 figures, and 11 data tables are released at:

<https://github.com/tanzimkhan24/genai-se-bangladesh>

under the MIT licence for code and CC BY 4.0 for the thesis. The random seed used throughout is SEED = 20260524.

Conflict of Interest

The author declares no competing interests.

Ethics

This study uses simulated data calibrated to publicly reported aggregate statistics. No human subjects research was conducted; no ethics approval was required.

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